

HW 5

1) a) From the zoo, we recognize this could be a uniform distribution, because the dice is fair and rolls integers.

$$E(X) = \frac{a+b}{2} = \frac{1+10}{2} = \boxed{11/2}$$

$$\text{Var}(X) = \frac{(b-a)(b-a+1)}{12} = \frac{(9)(10)}{12} = \frac{90}{12} = \boxed{\frac{33}{4}}$$

b) $E(Y) = \sum_{i=1}^8 E(X_i)$

$$= 8 \cdot \frac{11}{2} = \boxed{44}$$

$\text{Var}(Y) = \sum_{i=1}^8 \text{Var}(X_i)$ Because each dice roll is independent,

we can sum them together like

$$= 8 \cdot \left(\frac{33}{4}\right) = 2 \cdot 33 = \boxed{66}$$

$\sum_{i=1}^8 E(X_i)$ or $\sum_{i=1}^8 \text{Var}(X_i)$

2) Geometric distribution

Success probability = p $E[X] = \frac{1}{p}$
fail = $(1-p)$

expectation that
someone passes by and
joins the group

$E[X] = \frac{1}{p} + p(1) + (1-p)(1+E)$ Law of total probability;

$$E = \frac{1}{p} + (1-p)E$$

$\frac{1}{p}$ = find the first person,

$$E - (1-p)E = \frac{1}{p} + 1$$

$p(1)$ = find 1 more person (trio),

$$pE = 1 + p$$

$(1-p) \pm$ Failure, $(1+E)$ = start over,

$$E = \frac{1+p}{p}$$

when failure there is a new
Expectation. Then solve for E

Every λ minutes, new person. Multiply Every Expectation
by λ to get:

$$E[T] = \frac{1+p}{p} \cdot \lambda$$

$$E[T] = E[X] \cdot \lambda$$

Expected time found by
multiplying $E[X]$ by the
time between each person

3) $\binom{N}{2}$ photos 50ms

$$\binom{N}{2} = \frac{N^2 - N}{2} \cdot 50\text{ms} \rightarrow 25(N^2 - N) = \text{CPU Time}$$

If there are $\binom{N}{2}$ photos, each take 50ms,
the total time is $25(N^2 - N)$

$$25(E[N^2]) - E[N] = E[T] \quad \text{Linearity of Expectation}$$

$$\text{Var}(N) = E[N^2] - (E[N])^2 \rightarrow \text{Var}(N) = E[N^2] - (E[N])^2 \quad \text{Variance formula}$$

a) $E(N) = n$ $\text{Var}(N) = 0$ if N is a fixed n , then
 $E(T) = 25(n^2 - n)$ we simply plug n in for N , no variance

b) Binomial (n, p)

$$E[N] = np \quad \text{Var}(N) = np(1-p)$$

$$E[N^2] = np(1-p) + (np)^2 = np - np^2 + n^2 p^2$$

c) Poisson

$$E[N] = \lambda \quad \text{Var}(N) = \lambda$$

$$E[N^2] = \lambda + \lambda^2 = \lambda(\lambda + 1)$$

d) Bernoulli

$$E[N] = 7p + 20$$

$$\text{Var}(N) = 7^2 + \text{Var}(X) = 49p(1-p)$$

$$E[N^2] = 7^2(p-p^2) + (7p+20)^2$$

$$= 49(p-p^2) + (7p+20)^2$$

$$= 49p - 49p^2 + 49p^2 + 400 + 280p$$

$$= 329p + 400$$

4) a) Variables:

μ = # of customers

$\lambda = 3$ customers/hr

$P = \text{Profit} = \$5 + 2y$

$H = \text{total heart shaped cookies} = 2C$

$y = \text{cookie cost}$

Linearity of expectation

$$E[H] = E[2C] = 2E[C] = 2(24) = 48$$

$$\begin{aligned} 24 \text{ customers per day} &= \lambda \cdot 8 \text{ hr} \\ &= 3 \cdot 8 \\ &= 24 \end{aligned}$$

$$E[P] = E[(5 + 2y)C]$$

$$= (5 + 2y)E[C]$$

$$= 5 + 2y(24)$$

$$= \boxed{120 + 48y} \text{ Expected profit: } E[H] \cdot 5 + 2y$$

which is the profit

generated from the heart
cookies plus the expected
prof

$$b) E(P_{\text{normal}}) = 5 \cdot 24 = 120$$

$$\text{rate } \lambda = 3$$

$$= 3 \cdot 4 \text{ hrs} = 12 \text{ customers}$$

M = number of customers in 'new day'

$$P_{\text{new}} = 5 \cdot M$$

$$P(P_{\text{new}} \geq 120 | P_{\text{old}}) = P(L \geq 24)$$

The probability the profit on this new schedule is equal to the probability that the bakery gets more than 24 customers.

$$P(5M \geq 120)$$

$$P(M \geq 24)$$

$$\text{The opposite of } P(M \geq 24) = 1 - P(Y \leq 23)$$

$$P(\text{Profit} \geq 120) = 1 - F(23)$$

F is the cdf for the Poisson Random variable. $1 - F(23)$ is the chance that the amount of customers is greater than or equal to 24.

5) a) X is the # of correct problems

$n=20$ trials $p=0.85$

$X \sim (20, 0.85)$ Binomial Distribution

$$\begin{aligned} E[H_1] &= E\left[\frac{X}{20}\right] \quad H_1 = \frac{X}{20} \\ &= \frac{1}{20} E[X] \\ &= \frac{20 \cdot 0.85}{20} = 0.85 \end{aligned}$$

$$\text{Variance: } \text{Var}(H_1) = \text{Var}\left(\frac{X}{20}\right) = \frac{1}{20^2} \text{Var}(X)$$

$$= \frac{20 \cdot 0.85 \cdot 0.15}{400} = \frac{2.55}{400} = 0.006375$$

b) Average grade: H

Distribution:

$$H = \frac{1}{10} \sum_{i=1}^{10} H_i$$

We used iid, summation, because every question set is independent

$$E[H] = E[H_1] = 0.85$$

$$\text{Var}(H) = \text{Var}\left(\frac{1}{10} \sum_{i=1}^{10} H_i\right) = \frac{1}{100} \text{Var}\left(\sum_{i=1}^{10} H_i\right)$$

$$\text{Var}\left(\sum_{i=1}^{10} H_i\right) = \sum_{i=1}^{10} \text{Var}(H_i) = 10 \cdot 0.006375 = 0.006375$$

$$\text{Var}(H) = \frac{0.006375}{100} = \boxed{0.00006375}$$

c) I = indicator that problem is correct.

X_k = indicator that k is correct, $k=1,2,3$

$X_k \sim \text{Bernoulli}(0.6)$

$$P(S=1) = P(\text{success on attempt } 1,2,3)$$

$$= P(X_1=1) + P(X_1=0, P(X_2=1) + P(X_2=0, X_1=0, X_3=1))$$

Because they are independent, we just plug in:

$$= 0.6 + (0.4)(0.6) + (0.4)(0.4)(0.6)$$

$$= 0.6 + 0.24 + 0.096 = 0.936$$

$$E_1 = \frac{S}{20} \quad S \sim \text{Bernoulli}(0.936)$$

$$E[E_1] = E\left[\frac{S}{20}\right] = \frac{1}{20} E[S] = \frac{0.936}{20} = 0.0468$$

Variance:

$$\text{Var}(E_1) = \text{Var}\left(\frac{S}{20}\right)$$

$$= \frac{1}{400} \text{Var}(S)$$

$$= \frac{1}{400} (0.936)(0.064)$$

$$= \frac{0.059904}{400}$$

$$= 0.00014976$$

d) $E_i =$ grade on problem i

$$E = \sum_{i=1}^{20} E_i$$

$$E[E] = E\left[\sum_{i=1}^{20} E_i\right] = \sum_{i=1}^{20} E[E_i] = 20 \cdot 0.0468 \\ = 0.936$$

e) Part a has higher variances, this is because part b had more values comparing that when averaged out, makes the variance smaller. More extreme scores are "weighed" less when larger amounts of data is present.

b) a) Payout

Variables: T = time of touchdown

$T \sim \text{Geometric}(p)$

$$E[T] = 1/p$$

$$\text{Var}(T) = \frac{1-p}{p^2}$$

$$W = 150 - 10(T-1)$$

$$E[W] = 160 - 10E[T]$$

$$= 150 - (10T - 10)$$

$$= 160 - 10/p$$

$$= 160 - 10T$$

Variance:

$$\text{Var}(W) = \text{Var}(160 - 10T)$$

$$= (10)^2 \text{Var}(T)$$

$$= \frac{100(1-p)}{p^2}$$

b) G = single game payout

$$P(T \leq 15) = 1 - (1-p)^{15} \rightarrow 30$$

$$P(15 < T \leq 30) = (1-p)^{15} - (1-p)^{30} \rightarrow 10$$

$$P(T > 30) = (1-p)^{30} \rightarrow 0$$

$$E[G] = 30(1 - (1-p)^{15}) + 10((1-p)^{15} - (1-p)^{30})$$

$$= 30 - 20(1-p)^{15} - 10(1-p)^{30}$$

$$E[X] = 20 \cdot E[G]$$

$$= 600 - 400(1-p)^{15} - 200(1-p)^{30}$$